our limited experience. When we look a little further ahead we discover that the world is not as it appears to us: the Earth is round, and in Cape Town their feet are up and their heads are down. To trust immediate intuitions rather than collective examination that is rational, careful and intelligent is not wisdom: it is the presumption of an old man who refuses to believe that the great world outside his village is any different from the one which he has always known.

As vivid as it may appear to us, our experience of the passage of time does not need to reflect a fundamental aspect of reality. But if it is not fundamental, where does it come from, our vivid experience of the passage of time?

I think that the answer lies in the intimate connection between time and heat. There is a detectable difference between the past and the future only when there is flow of heat. Heat is linked to probability; and probability in turn is linked to the fact that our interactions with the rest of the world do not register the fine details of reality. The flow of time emerges thus from physics, but not in the context of an exact description of things as they are. It emerges, rather, in the context of statistics and of thermodynamics. This may hold the key to the enigma of time. The ‘present’ does not exist in an objective sense any more than ‘here’ exists objectively, but the microscopic interactions within the world prompt the emergence of temporal phenomena within a system (for instance, ourselves) which only interacts through the medium of a myriad of variables.

Our memory and our consciousness are built on these statistical phenomena. For a hypothetically supersensible being there would be no ‘flowing’ of time: the universe would be a single block of past, present and future. But due to the limitations of our consciousness we only perceive a blurred vision of the world, and live in time. Borrowing words from my Italian editor, ‘what’s non-apparent is much vaster than what’s apparent’. From this limited, blurred focus we get our perception of the passage of time. Is that clear? No, it isn’t. There is so much still to be understood.

Time sits at the centre of the tangle of problems raised by the intersection of gravity, quantum mechanics and thermodynamics. A tangle of problems where we are still in the dark. If there is something which we are perhaps beginning to understand about quantum gravity, which combines two of the three pieces of the puzzle, we do not yet have a theory capable of drawing together all three pieces of our fundamental knowledge of the world.